

NIBSS by Phoenix

API Technical Documentation

TS Academy — Phoenix Cohort

Built By

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1. Introduction

NIBSS by Phoenix is a simulated Nigerian Interbank Settlement System (NIBSS) API built by TS Academy's Phoenix cohort as a hands-on training exercise. It mirrors the architecture and flows of real-world Nigerian fintech infrastructure, giving students practical experience with fintech API design, identity management, account operations, and interbank transfers.

This document provides complete technical documentation for all endpoints in the collection, describing request structures, required fields, authentication requirements, and the intended operational flow.

2. Project Team

Field	Type	Required	Description
Onyekachi Obute	Tutor	—	Lead instructor and API architect for the Phoenix cohort project
John Oloruntobi	Student Coordinator	—	Coordinated student tasks and integration testing
Jude Ani	Student	—	Endpoint development and Postman collection authoring

3. Base Configuration

3.1 Base URL

All endpoints are relative to the configured base URL. In the Postman collection, this is stored as the environment variable:

```
https://nibssbyphoenix.onrender.com/
```

3.2 Authentication

Most endpoints (except Fintech Onboarding, Login, BVN and NIN) require a Bearer Token passed in the Authorization header. The token is a JSON Web Token (JWT) returned after a successful login.

```
Authorization: Bearer <your_jwt_token>
```

The JWT payload includes the following claims:

Field	Type	Required	Description
fintechId	string	—	Unique identifier for the onboarded fintech

name	string	—	Name of the fintech or user
email	string	—	Email address
bankCode	string	—	Assigned bank code (e.g. 108)
bankName	string	—	Assigned bank name (e.g. KAC Bank)
iat	number	—	Issued-at timestamp (Unix)
exp	number	—	Expiry timestamp (Unix)

4. API Flow Overview

The NIBSS by Phoenix API is designed around a structured lifecycle. Understanding the intended sequence of calls is essential for correct integration. The flow is divided into four stages:

01	Identity Setup	Register BVN and NIN records into the NIBSS identity store. These represent the central identity layer that all banks and fintechs can query.
02	Fintech Onboarding & Auth	Onboard a fintech institution and obtain a JWT access token via login. This token authenticates all subsequent API calls.
03	Account Operations	Create bank accounts for customers, retrieve accounts, check balances, and perform name enquiries on account numbers.
04	Transactions	Execute interbank fund transfers using account numbers and track transaction status using a unique Transaction ID (TSQ).

The diagram below illustrates the typical call sequence for a complete user journey — from identity registration through a successful interbank transfer:

1	BANK Admin registers BVN & NIN	POST /api/insertBvn → POST /api/insertNin
2	Fintech onboards to the system	POST /api/fintech/onboard
3	Fintech logs in → receives JWT token	POST /api/auth/token
4	Fintech creates customer accounts	POST /api/account/create
5	Sender checks recipient name	GET /api/account/name-enquiry/{accountNo}
6	Bank verifies BVN or NIN	POST /api/validateBvn POST /api/validateNin
7	Fund transfer is initiated	POST /api/transfer
8	Transaction status is queried (TSQ)	GET /api/transaction/{transactionId}
9	Account balances are verified	GET /api/account/balance/{accountNo}

5. API Endpoint Reference

5.1 Fintech Onboarding

Registers a new fintech institution on the NIBSS platform. Upon successful registration, the system auto-assigns a unique Bank Code and Bank Name to the fintech. No authentication token is required for this endpoint.

POST`/api/fintech/onboard`*Register a new fintech institution*

Request Body

Field	Type	Required	Description
name	string	Yes	The registered name of the fintech institution or representative
email	string	Yes	Contact email address — must be unique per fintech

Sample Request

```
{
  "name": "Phoenix Bank",
  "email": "phoenix@tsacademyonline.com"
}
```

Sample Response (201 Created)

```
{
  "apiKey": "8afa6eec4f941ee25c7853f451d32f88",
  "apiSecret": "de6072722c2dde5906f6d2a6c5f45dae2eacbaa90f31d3f37691ea8993443367",
  "bankCode": "260",
  "bankName": "PHC Bank"
}
```

Notes

The bankCode and bankName are automatically generated by the system and are embedded in the JWT token returned at login. These are used to identify the fintech across all interbank transactions.

5.2 Login

Authenticates an already-onboarded fintech and returns a signed JWT access token. This token expires after 1 hour and must be included in the Authorization header of all subsequent protected endpoints.

POST`/api/auth/token`*Authenticate and receive a JWT token*

Request Body

Field	Type	Required	Description
apikey	string	Yes	Api key generated during fintech onboarding
apisecret	string	Yes	Api Secret generated during fintech onboarding

Sample Request

```
{
  "apiKey": "8afa6eec4f94ee25c7853f451d32f88",
  "apiSecret": "de6072722c2dde06f6d2a6c5f45dae2eacbaa90f31d3f37691ea8993443367"
}
```

Sample Response (200 OK)

```
{
  "token":
"eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJmaW50ZWNoSWQiOiI2OWUxNGVhOWQ4MjYxN2Q0MjA4YTU1YWEiLCJuaW11IjoIQUNDRVNTIiwiaWZlhaWwiOiJvY29idXRlQGdtYWlsLmNvbSI6ImJhbmtDb2RlIjoIMjYwIiwiaWF0IjE5bWUuOjBQ0MgQmFuayIsImhhdCI6MTc3NjQyODM3OSwiZXhwIjozNzc2NDMxOTc5fQ.95ocxEtfxRd9743GUn2sySA8D2c3ojtcNG4EVuDsFvo",
  "fintech": {
    "name": "Phoenix Bank",
    "email": "phoenix@tsacademyonline.com",
    "bankCode": "260",
    "bankName": "PHC Bank"
  }
}
```

Notes

The token encodes fintechId, name, email, bankCode, and bankName. It is valid for 1 hour (iat to exp difference is 3600 seconds). All protected routes will return 401 Unauthorized if a valid token is not provided.

5.3 Create Account

Creates a new bank account for a customer under the authenticated fintech's bank. The system auto-generates a 10-digit NUBAN-style account number and links it to the fintech's assigned bank code.

POST`/api/account/create`*Create a new customer bank account*

Authentication

Bearer Token required.

Request Body

Field	Type	Required	Description
<code>kycType</code>	string	Yes	Either BVN or NIN (depending on what you want to create the account with)
<code>kycID</code>	string	Yes	Nin or bvn number
<code>dob</code>	string	Yes	Date of birth as it in the bvn

Sample Request

```
{
  "kycType": "bvn",
  "kycID": "10840712847",
  "dob": "2005-04-04"
}
```

Sample Response (201 Created)

```
{
  "message": "Account created successfully",
  "accountNumber": "1084071287",
  "bankCode": "108",
  "bankName": "KAC Bank 9263",
  "balance": 15000
}
```

```
}
```

Error Response (if the bvn has been used to create account before)

```
{  
  "message": "bvn already linked to an account"  
}
```

5.4 Name Enquiry

Resolves an account number to the account holder's registered name. This is the standard pre-transfer verification step — a fintech must always confirm the recipient name before initiating a transfer.

GET`/api/account/name-
enquiry/{accountNumber}`*Resolve account number to account holder name*

Authentication

Bearer Token required.

Path Parameter

Field	Type	Required	Description
accountNumber	string	Yes	The 10-digit NUBAN account number to look up (e.g. 1084071287)

Sample Response (200 OK)

```
{  
  "accountNumber": "1084071287",  
  "accountName": "John Doe",  
  "bankName": "Phc bank"  
}
```

5.5 Get All Accounts

Returns a list of all bank accounts associated with the authenticated fintech's bank. Useful for administrative oversight and reporting.

GET`/api/accounts`*Retrieve all accounts under the authenticated fintech*

Authentication

Bearer Token required.

Request Body

No request body required.

Sample Response (200 OK)

```
{
  "accounts": [
    {
      "accountNumber": "1084071287",
      "accountName": "Onyekachi Obute",
      "balance": 500000
    },
    ...
  ]
}
```

5.6 Account Balance

Retrieves the current ledger balance for a specific account number. The account must belong to the authenticated fintech's bank.

GET`/api/account/balance/{accountNumber}`*Get balance for a specific account*

Authentication

Bearer Token required.

Path Parameter

Field	Type	Required	Description
accountNumber	string	Yes	The 10-digit NUBAN account number (e.g. 1084071287)

Sample Response (200 OK)

```
{
```

```
"accountNumber": "1084071287",  
"balance": 420000  
}
```

5.7 Transfer

Initiates an interbank fund transfer between two accounts. The transfer is debited from the sender's account and credited to the recipient's account. If both accounts belong to different bank codes, the transfer is routed through the NIBSS settlement layer. The system returns a unique Transaction ID (TSQ) that can be used to query transaction status.

POST`/api/transfer`*Initiate an interbank fund transfer*

Authentication

Bearer Token required.

Request Body

Field	Type	Required	Description
from	string	Yes	Sender's 10-digit NUBAN account number
to	string	Yes	Recipient's 10-digit NUBAN account number
amount	string	Yes	Amount to transfer (in Naira)

Sample Request

```
{
  "from": "1084071287",
  "to": "1087207670",
  "amount": "80000"
}
```

Sample Response (200 OK)

```
{
  "message": "Transfer successful",
  "transactionId": "TX1776340463722",
  "amount": 80000,
  "from": "1084071287",
  "to": "1087207670",
  "status": "SUCCESS"
}
```

Notes

Always perform a Name Enquiry (Section 5.4) and identity validation (Section 5.8 or 5.9) before initiating a transfer. Insufficient funds will return a 400 Bad Request error.

5.8 Transaction Status Query (TSQ)

Queries the status of a previously initiated transfer using the unique Transaction ID returned by the Transfer endpoint. This is used to confirm whether a transaction completed successfully, is pending, or failed — particularly important for resolving interbank transfer delays.

GET`/api/transaction/{transactionId}`*Query the status of a specific transaction*

Authentication

Bearer Token required.

Path Parameter

Field	Type	Required	Description
transactionId	string	Yes	The unique transaction ID returned from the Transfer endpoint (e.g. TX1776340463722)

Sample Response (200 OK)

```
{
  "transactionId": "TX1776340463722",
  "status": "SUCCESS",
  "amount": 80000,
  "from": "1084071287",
  "to": "1087207670",
  "timestamp": "2026-04-16T10:31:03Z"
}
```

5.9 Create BVN

Registers a new Bank Verification Number (BVN) record in the NIBSS central identity store. This is an administrative operation typically performed by NIBSS operators or for seeding test data. The BVN is linked to a person's biographic information and serves as the single identity layer across all Nigerian banks.

POST`/api/insertBvn`*Register a new BVN record in the NIBSS identity store*

Authentication

Bearer Token required.

Request Body

Field	Type	Required	Description
bvn	string	Yes	11-digit BVN number to register (e.g. 10840712847)
firstName	string	Yes	First name of the BVN holder
lastName	string	Yes	Last name of the BVN holder
dob	string	Yes	Date of birth in YYYY-MM-DD format (e.g. 2005-04-04)
phone	string	Yes	Phone number associated with the BVN (e.g. 07869584455)

Sample Request

```
{
  "bvn": "10840712847",
  "firstName": "Onyekachi",
  "lastName": "Obute",
  "dob": "2005-04-04",
  "phone": "07869584455"
}
```

Sample Response (201 Created)

```
{
  "message": "BVN record created successfully",
  "bvn": "10840712847"
}
```

5.10 Create NIN

Registers a new National Identification Number (NIN) record into the NIBSS identity store. NIN is issued by NIMC (National Identity Management Commission) and is used alongside BVN for multi-factor identity verification in Nigerian banking.

POST`/api/insertNin`*Register a new NIN record in the NIBSS identity store*

Authentication

Bearer Token required.

Request Body

Field	Type	Required	Description
nin	string	Yes	11-digit NIN number to register (e.g. 10840712847)
firstName	string	Yes	First name matching NIMC records
lastName	string	Yes	Last name matching NIMC records
dob	string	Yes	Date of birth in YYYY-MM-DD format

Sample Request

```
{
  "nin": "10840712847",
  "firstName": "Onyekachi",
  "lastName": "Obute",
  "dob": "2005-04-04"
}
```

Sample Response (201 Created)

```
{
  "message": "NIN record created successfully",
  "nin": "10840712847"
}
```

5.11 Validate BVN

Verifies that a given BVN exists in the NIBSS identity store. Used by fintechs to confirm a customer's identity before account creation or high-value transactions. Returns the associated biographic details for cross-checking.

POST`/api/validateBvn`*Validate a BVN against the NIBSS identity store*

Authentication

Bearer Token required.

Request Body

Field	Type	Required	Description
bvn	string	Yes	The 11-digit BVN to validate

Sample Request

```
{
  "bvn": "10840712847"
}
```

Sample Response (200 OK)

```
{
  "valid": true,
  "bvn": "10840712847",
  "firstName": "Onyekachi",
  "lastName": "Obute",
  "dob": "2005-04-04"
}
```

5.12 Validate NIN

Verifies that a given NIN exists in the NIBSS identity store and returns the associated identity information. Used alongside BVN validation to provide multi-layered Know Your Customer (KYC) checks.

POST`/api/validateNin`*Validate a NIN against the NIBSS identity store*

Authentication

Bearer Token required.

Request Body

Field	Type	Required	Description
nin	string	Yes	The 11-digit NIN to validate

Sample Request

```
{
  "nin": "10840712847"
}
```

Sample Response (200 OK)

```
{
  "valid": true,
  "nin": "10840712847",
  "firstName": "Onyekachi",
  "lastName": "Obute",
  "dob": "2005-04-04"
}
```

6. HTTP Status Codes

The following standard HTTP response codes are used across all endpoints:

Code	Status	Description
200	200 OK	Request succeeded. Data returned as expected.
201	201 Created	Resource created successfully (e.g. new account, BVN, NIN).
400	400 Bad Request	Invalid input or missing required fields. Check request body.
401	401 Unauthorized	Missing or expired JWT token. Re-authenticate to get a fresh token.
404	404 Not Found	Requested resource does not exist (e.g. account not found).
409	409 Conflict	Duplicate resource (e.g. BVN or NIN already registered).
500	500 Internal Server Error	Server-side error. Report to the development team.

7. Endpoint Summary

#	Method	Endpoint	Auth	Purpose
1	POST	/api/fintech/onboard	None	Fintech Onboarding
2	POST	/api/auth/token	None	Login
3	POST	/api/account/create	JWT	Create Account
4	GET	/api/account/name-enquiry/{no}	JWT	Name Enquiry
5	GET	/api/accounts	JWT	Get All Accounts
6	GET	/api/account/balance/{no}	JWT	Account Balance
7	POST	/api/transfer	JWT	Fund Transfer
8	GET	/api/transaction/{id}	JWT	TSQ — Transaction Status
9	POST	/api/insertBvn	None	Create BVN Record
10	POST	/api/insertNin	None	Create NIN Record
11	POST	/api/validateBvn	None	Validate BVN
12	POST	/api/validateNin	None	Validate NIN

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